

Pulmonary Hypertension and Pulmonary Circulation

Pulmonary hypertension (BMPR2 and related genes, SOX17, HLA-DPA1/DPB1), pulmonary embolism and thromboembolic pulmonary hypertension, lung and hypoxia (PiMZ), and right ventricular function

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Favourable. No pathogenic variant was detected in the genes of arterial pulmonary hypertension that the chip reads safely: ACVRL1 is normal. Attention: of the ~1970 pathogenic variants known in BMPR2 and in more 11 related genes, the chip only covers a small fraction, so the BMPR2 is not evaluated by this test. In **SOX17 rs10103692**, the main common locus of pulmonary hypertension, Sérgio F is G/A: it has less risk alleles than most people, which is G/G. It also has no risk alleles of PDE1A or CBLN2. Factor V Leiden and prothrombin G20210A are negative, the group is O (ABO rs505922 T/T, with less factor VIII) and does not have PiS. Pulmonary embolism is low in a genomic score (percentile 1.4) and typical in another (25.1). The function of the right ventricle is typical in the loci (47.1).

Unfavourable. Venous thromboembolism is high: scores at the 86.3 and 84.4 percentiles (§49) and loci load at percentile 95.1. The load of pulmonary embolism loci is at 90.8. F11 rs2289252 is T/T (2 alleles; locus of thrombosis and HPTEC) and **FGG** has 1 risk allele. On the lung side, the alpha-1 antitrypsin PiMZ and the emphysema score at percentile 96.2 (§75) are the most likely pathway for secondary pulmonary hypertension, but only with tobacco or tobacco.

What does it mean? Primary pulmonary hypertension (1 group) is rare (~1-2 by 100 000 per year). Nothing here suggests a hereditary risk, but BMPR2 remains unevaluated. The most plausible pathways for Sérgio F are the secondary ones: thromboembolic (by genetic predisposition to thrombosis) and pulmonary (PiMZ, only with tobacco or tobacco). Hence the usefulness of preventing thrombosis, never smoking and watching unexplained air loss. This report complements Lung (§75) and Coagulation (§49).

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality DR2 ≥ 0.8 . Under this threshold SNP is treated as SNP not found in the data.
- **Key variants** 13 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 11 of the chip, 2 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 1 SNPs needed conversion to the complementary strand and 1 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 6 PGS Catalog scores calculated locally with 1,036,984 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population (n = 633) calculated with exactly the same variants. Percentage ≥ 80 : MORE LIKELY/HIGH; ≤ 20 : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population (n = 633) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 360 loci, 305 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

In all these scores, a higher percentile = more risk (pulmonary embolism, venous thrombosis, emphysema). The venous thrombosis scores are those of \$49 and the emphysema score is that of \$75, with the same values. There are no published polygenic scores for arterial pulmonary hypertension.

Pulmonary embolism and thrombosis

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Pulmonary embolism (I26)	PGS004460 (Jung 2024, LDpred2, 1.06 million variants)	1,036,079 1,059,939 (weighted coverage 97%)	1.4	LESS LIKELY	favourable
Pulmonary embolism	PGS003861 (Zhang 2023, PRSice-2, 288 variants)	283 288 (weighted coverage 98%)	25.1	TYPICAL	typical
Coagular in the lung	PGS001279 (Tanigawa 2022, 94 variants)	73 94 (weighted coverage 84%)	67.6	TYPICAL	typical
Venous thrombosis	PGS000043 (Klarin 2019, 297 variants)	265 297 (weighted coverage 84%)	86.3	MORE LIKELY	unfavourable
Venous thrombosis	PGS003456 (recent score, 273 loci)	255 273 (weighted coverage 88%)	84.4	MORE LIKELY	unfavourable

- **Venous Thromboembolism:** Already in \$49.
- **Venous Thromboembolism:** Already in \$49.

Lung (3 group)

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Emphysema	PGS001326 (Tanigawa 2022, 42 variants)	29 42 (weighted coverage 73%)	96.2	MORE LIKELY	unfavourable

- **Emphasis:** In \$75 (it probably includes the effect of PiZ).

4. Key variants by topic

Common loci of arterial pulmonary hypertension, the only pathogenic variant of the genes of the BMP pathway read safely (ACVRL1), thrombosis factors (§49) and alpha-1 antitrypsin (§75).

Pulmonary hypertension

Hereditary pulmonary hypertension is mainly due to rare variants of BMPR2 and genes from the same pathway BMP/TGF- β (ACVRL1, ENG, SMAD9, GDF2, SOX17, TBX4, KCNK3, EIF2AK4). Common loci (SOX17, HLA-DPA1/DPB1) change little the risk.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
SOX17	rs10103692		G/A	imputation DR2=0.99	G (1 copy/s)	unfavourable (light)	Enhancer of SOX17 in the pulmonary endothelium: the G allele (common, ~90%) is associated with arterial pulmonary hypertension (OR ~1.8)	Moderate
SOX17	rs13266183		C/C	chip	C (2 copy/s)	unfavourable	Second sign of SOX17 in arterial pulmonary hypertension	Moderate
HLA-DPA1/DPB1	rs2856830		T/C	chip	C (1 copy/s)	without judgment	LOCUS HLA of arterial pulmonary hypertension; the homozygous C/C have better survival in patients	Moderate
PDE1A/DNAJC10	rs71427857		G/G	chip	A (0 copy/s)	favourable	Rare locus of pulmonary hypertension (GWAS)	Weak
CBLN2	rs2217560		A/A	imputation DR2=0.98	G (0 copy/s)	favourable	Arterial pulmonary arterial hypertension locus without BMPR2 mutation	Weak
ACVRL1 (ALK1)	rs139142865		C/C	chip	A (0 copy/s)	favourable	Pathogenic variant of ACVRL1 (heritable hemorrhagic telangiectasia and arterial pulmonary hypertension); the only pathogenic variant of these genes safely read by the chip	Strong

Thrombosis and pulmonary embolism

Unresolved pulmonary embolism may progress to chronic thromboembolic pulmonary hypertension (HPTEC). F11, ABO and fibrinogen are loci shared between thrombosis and HPTEC.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
F5	rs6025	Leiden	C/C	chip	T (0 copy/s)	favourable	Factor V Leiden: venous thromboembolism ~3-5x in heterozygous (§49)	Strong
F2	rs1799963	G20210A	G/G	chip	A (0 copy/s)	favourable	Prothrombin G20210A: venous thromboembolism ~2-3x (§49)	Strong
F11	rs2289252		T/T	chip	T (2 copy/s)	unfavourable	The T allele increases the XI factor; venous thromboembolism and chronic thromboembolic pulmonary hypertension (GWAS)	Strong
ABO	rs505922	non-O	T/T	chip	C (0 copy/s)	favourable	The C allele marks the non-O groups (plus factor VIII and von Willebrand; plus thrombosis and thromboembolic pulmonary hypertension)	Moderate

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
FGG	rs2066865		G/A	chip	A (1 copy/s)	unfavourable (light)	Allele A reduces fibrinogen γ' ; venous thromboembolism $\sim 1.2\times$ per copy (\$49)	Moderate

Lung and hypoxia

Chronic lung disease and hypoxia contract pulmonary arterioles (3 group pulmonary hypertension). The alpha-1 antitrypsin PiMZ increases the risk of emphysema with tobacco or tobacco.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
SERPINA1	rs28929474	PiZ	C/T	chip	T (1 copy/s)	unfavourable	PiZ: with tobacco or tobacco, emphysema/COPD, which is a cause of pulmonary hypertension in the 3 group (\$75)	Strong
SERPINA1	rs17580	PiS	T/T	chip	A (0 copy/s)	favourable	PiS: slight deficiency; relevant combined with Z	Strong

5. GWAS loci burden

Sum of GWAS Catalog loci ($p \leq 510-8$) of pulmonary embolism, venous thromboembolism and ejection fraction of the right ventricle (here, a high percentile = better). The published pulmonary hypertension loci are few and almost all without indicated risk allele; they were read individually.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
Pulmonary embolism	44 (37)	41 of 74	37	90.8	unfavourable
Venous thromboembolism	297 (250)	237 of 500	246	95.1	unfavourable
Right ventricle (ejection fraction)	19 (18)	19 of 36	18	47.1	typical

6. Favourable

- **No pathogenic variant detected** in genes of the BMP pathway read safely (ACVRL1 rs139142865 C/C). Strong for the tested variant; BMPR2 is not evaluated by the chip.
- **SOX17 rs10103692 G/A**: less risk alleles than most (G/G). **PDE1A G/G** and **CBLN2 A/A** without risk alleles. **Weak moderate.**
- **Factor V Leiden C/C and prothrombin G20210A G/G** negative; group O (ABO rs505922 T/T). **Strong.**
- **Pulmonary embolism**: genomic score at percentile 1.4 and 288 variant score in the 25.1. **Moderate.**
- **Absent PiS** (T/T). **Strong.**
- **Right ventricle**: load of typical ejection fraction (47.1). **Weak.**

7. Unfavourable

- **Venous thromboembolism:** scores at the 86.3 and 84.4 percentiles (§49) and loci load at percentile 95.1. **Moderate.**
- **Pulmonary embolism:** loci load at 90.8; the score of 94 variants "lung clot" is at 67.6. The signs of pulmonary embolism disagree with each other (low or typical scores, high loci). **Weak moderate.**
- **F11 rs2289252 T/T** (2 alleles with more factor XI; locus of thrombosis and HPTEC) and **FGG rs2066865 G/A.** **Moderate.**
- **SERPINA1 PiMZ** with emphysema at percentile 96.2 (§75): route for pulmonary hypertension of the 3 group, only with tobacco or tobacco. **Moderate.**
- **SOX17 rs13266183 C/C:** 2 copies of the associated common allele (~73% of alleles in the population). **Weak.**

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Vascular Aging	Pulmonary arterioles remodeling and risk of venous thrombosis (scores at the 86.3 and 84.4 percentiles)	Unfavourable (mild)
Intercellular Communication	Via BMP/TGF- β of the pulmonary endothelium (BMPR2, ACVRL1, SOX17): no variants detected in the read positions	Neutral
Remodelling of the extracellular matrix	Pulmonary hypertension involves thickening and fibrosis of the pulmonary arteries	Neutral
Fragmentation of Elastin	PiMZ and emphysema at percentile 96.2: loss of lung elastin with tobacco or tobacco	Unfavourable (conditional)
Chronic Inflammation	Perivascular inflammation and autoimmunity (HLA-DPA1/DPB1) participate in arterial pulmonary hypertension	Neutral
Mitochondrial dysfunction	Pulmonary artery cells in arterial pulmonary hypertension change to glycolysis (warburg effect); without genetic marker here	Neutral

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Lack of unexplained air**, easy tiredness, exhaustion, chest pain or swelling of the ankles that do not improve: ask the doctor for an echocardiogram with evaluation of the right ventricle and pulmonary artery. *Suggestion (not applied).*
2. **After any pulmonary embolism**: reevaluate at 3–6 months, with echocardiogram and, if there is a lack of air, ventilation-perfusion scintigraphy, to exclude HPTEC. It is the form of pulmonary hypertension that can be cured. *Suggestion (not applied).*
3. **Preventing thrombosis** (as in §49): early mobilization and prophylaxis in surgeries (anesthesia and surgery report), compression and hydration on flights of more than 4 hours, standing up and walking every 1–2 hours and avoiding dehydration. *Suggestion (not applied).*
4. **Avoiding drugs and drugs associated with pulmonary hypertension**: amphetamines and methamphetamine, cocaine, fenfluramine/benfluorex/aminorex appetite suppressors, dasatinib and interferon. Inform the doctor if any is proposed. *Suggestion (not applied).*
5. **Zero tobacco, vape and tobacco** (PiMZ + high genetic emphysema): is the measure that most protects against pulmonary hypertension in the 3 group. *Suggestion (not applied).*
6. **Altitude**: in stays above 2500 m, gradually rise and descend if there is a lack of disproportionate air at rest, cough or extreme tiredness. The low genetic hemoglobin is in §92. *Suggestion (not applied).*
7. **Sleep**: if they appear high snoring, breathing pauses or daytime sleepiness, do a sleep study, because untreated apnea causes pulmonary hypertension. Genetic apnea is typical (§41). *Suggestion (not applied).*
8. **Signs of hereditary hemorrhagic telangiectasia** (repeated nasal hemorrhages, small red vessels on the lips, tongue or fingers), or family history of pulmonary arterial hypertension or sudden death: ask for a clinical panel of BMPR2, ACVRL1, ENG, SMAD9, GDF2, SOX17 and TBX4, because the chip does not evaluate them. *Suggestion (not applied).*
9. **Iron**: keeping ferritin below 100 (plan) but avoiding iron deficiency, which worsens pulmonary circulation and tolerance to altitude. *Suggestion (not applied).*
10. **Information**: tadalafil 5 mg belongs to the class used in the treatment of arterial pulmonary hypertension (at a higher dose); there is no evidence that it prevents it. *Suggestion (not applied).*

10. Limitations

- **BMPR2 is not evaluated by this chip:** of the ~1970 pathogenic or probably pathogenic variants of 12 genes of arterial pulmonary hypertension and hereditary hemorrhagic telangiectasia in the ClinVar, only one fraction is in the chip, and only one has been dealt with safely. Exclusion requires clinical sequencing.
- The common loci of pulmonary arterial hypertension have modest effects and were discovered in patients; they do not serve to predict the risk in the population.
- The signs of pulmonary embolism disagree (low or typical scores and high loci load). The venous thrombosis score (§49) and F11 T/T are the most consistent signs.
- The emphysema score has few variants and includes the effect of PiZ.
- The load of loci GWAS sums SNPs in linkage imbalance; it is descriptive.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The published allele of effect is compared with the alleles of the imputed VCF; in the palindromic SNPs the strand + (standard of the GWAS Catalog and the PGS Catalog) is assumed.
- **Rare genes:** The pathogenic and probably pathogenic variants of ClinVar in BMP2, ACVRL1, ENG, SMAD9, CAV1, KCNK3, TBX4, SOX17, GDF2, EIF2AK4, ATP13A3 and AQP1 were crossed with the chip positions.
- **Polygenic scores:** Harmonized GRCh38 files from PGS Catalog (6 scores: 3 of pulmonary embolism, 2 of venous thrombosis and 1 of emphysema). Score = Δ coefficient $\times$ dosage. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog ($p \leq 5 \times 10^{-8}$), alleles guided by the published effect signal, and empirical calibration in the same reference population.
- **Interpretation:** Rhodes 2019 (GWAS of arterial pulmonary hypertension: SOX17 and HLA-DPA1/DPB1), Gräf 2018 and Zhu 2019 (rare genes), Humbert 2022 (pulmonary hypertension orientations), Liley 2020 (HPTEC), Klarin 2019 and Lindström 2019 (thrombosis), GWAS Catalog and ClinVar. The complete references are available on request.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Pulmonary hypertension	SOX17	rs10103692	G/A	imputation DR2=0.99	G	1	strand +	1 copy of risk allele
Pulmonary hypertension	SOX17	rs13266183	C/C	chip	C	2	strand +	2 risk allele copies
Pulmonary hypertension	HLA-DPA1/ DPB1	rs2856830	T/C	chip	C	1	strand +	1 copy of effect allele
Pulmonary hypertension	PDE1A/ DNAJC10	rs71427857	G/G	chip	A	0	strand +	without the risk allele
Pulmonary hypertension	CBLN2	rs2217560	A/A	imputation DR2=0.98	G	0	strand +	without the risk allele
Pulmonary hypertension	ACVRL1 (ALK1)	rs139142865	C/C	chip	A	0	complement (strand)	without the risk allele
Thrombosis and pulmonary embolism	F5	rs6025	C/C	chip	T	0	strand +	without the risk allele
Thrombosis and pulmonary embolism	F2	rs1799963	G/G	chip	A	0	strand +	without the risk allele
Thrombosis and pulmonary embolism	F11	rs2289252	T/T	chip	T	2	strand +	2 risk allele copies
Thrombosis and pulmonary embolism	ABO	rs505922	T/T	chip	C	0	strand +	without the risk allele
Thrombosis and pulmonary embolism	FGG	rs2066865	G/A	chip	A	1	strand +	1 copy of risk allele
Lung and hypoxia	SERPINA1	rs28929474	C/T	chip	T	1	strand +	PiMZ: alpha-1 antitrypsin ~60%; pulmonary risk mainly with tobacco or tobacco
Lung and hypoxia	SERPINA1	rs17580	T/T	chip	A	0	strand + (palindromic; assumed +)	without the risk allele